

The EPFL logo is displayed in a bold, red, sans-serif font in the top left corner. The background of the slide is a photograph of a modern building with a large, white, angular roof and a glass facade, set against a clear blue sky. A person is blurred in the foreground on the left, suggesting movement.

EPFL

CREATIVITY

Technology Ventures

■ École
polytechnique
fédérale
de Lausanne

LECTURER

Dr. Davide Bavato
College of Management of
Technology

Recap

Date	Topics	Guests
Sep 9	1. Introduction to entrepreneurship	
Sep 16	2. Clinic: ideation and creativity	
Sep 23	3. Opportunities from technology	
Sep 30	4. Opportunities for society	
Oct 7	5. Presentations: opportunity set	Gallo & Conti (Startup Launchpad), Lattuada (Technology Transfer Office), Delisle (SPOT)
Oct 14	6. Validation I: scientific approach	
Oct 21	--- no session ---	
Oct 28	7. Keynote: <i>The uncomfortable truth</i>	Vincent Bieri (Nextthink co-founder )
Nov 4	8. Validation II: industry analysis	
Nov 11	9. Feedback session: validation review	Student founders (tbc)
Nov 18	10. Business models I: creating value	
Nov 25	11. Field trip: EPFL Investor Day	
Dec 2	12. Business models II: capturing value	
Dec 9	13. Clinic: pitching and deck design	
Dec 16	14. Presentation: venture concepts	Scetta (MassChallenge), Molina (TalentKick), Lamotte (InnovationPark), others tbc

Special schedule:
17:30 – 19:00
CM 11 

Creativity clinic

- The science of creativity: A primer
- Individual creativity: insight & ideation
- Collective creativity + *introduce & connect*
- Human-AI co-creation
- Next steps – *view on first assignment*

PART I

The science of creativity

From creative genius to everyday insight

A response considered to be *novel*, *unusual*, *surprising* – and at the same time *relevant* to the task, situation, audience



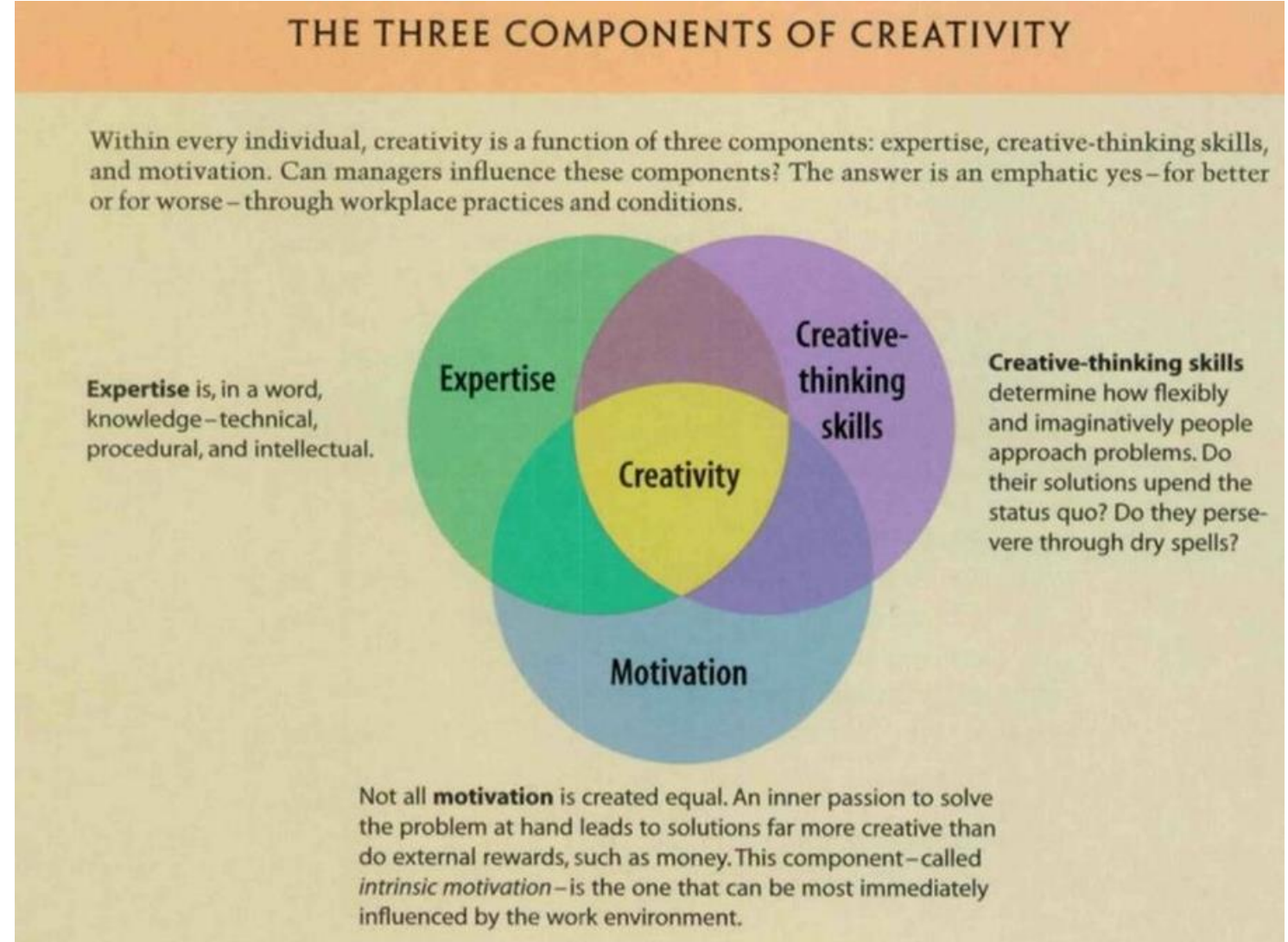
From individual trait to context-specific performance

Individuals can influence their own (and others) performance



Componential theory of creativity

- **Domain-relevant expertise** (technical skills, formal and informal knowledge)
- **Task motivation** (attitude, intrinsic motivation, external rewards)
- **Creative-thinking skills** (incl. knowledge of idea generation heuristics)



Creativity & e'ship

- **Creative destruction (Schumpeter).** *“The function of entrepreneurs is to reform or revolutionize the pattern of production by exploiting an invention or, more generally, an untried technological possibility for producing a new commodity or producing an old one in a new way, by opening up a new source of supply of materials or a new outlet for products, by reorganizing an industry and so on.”*

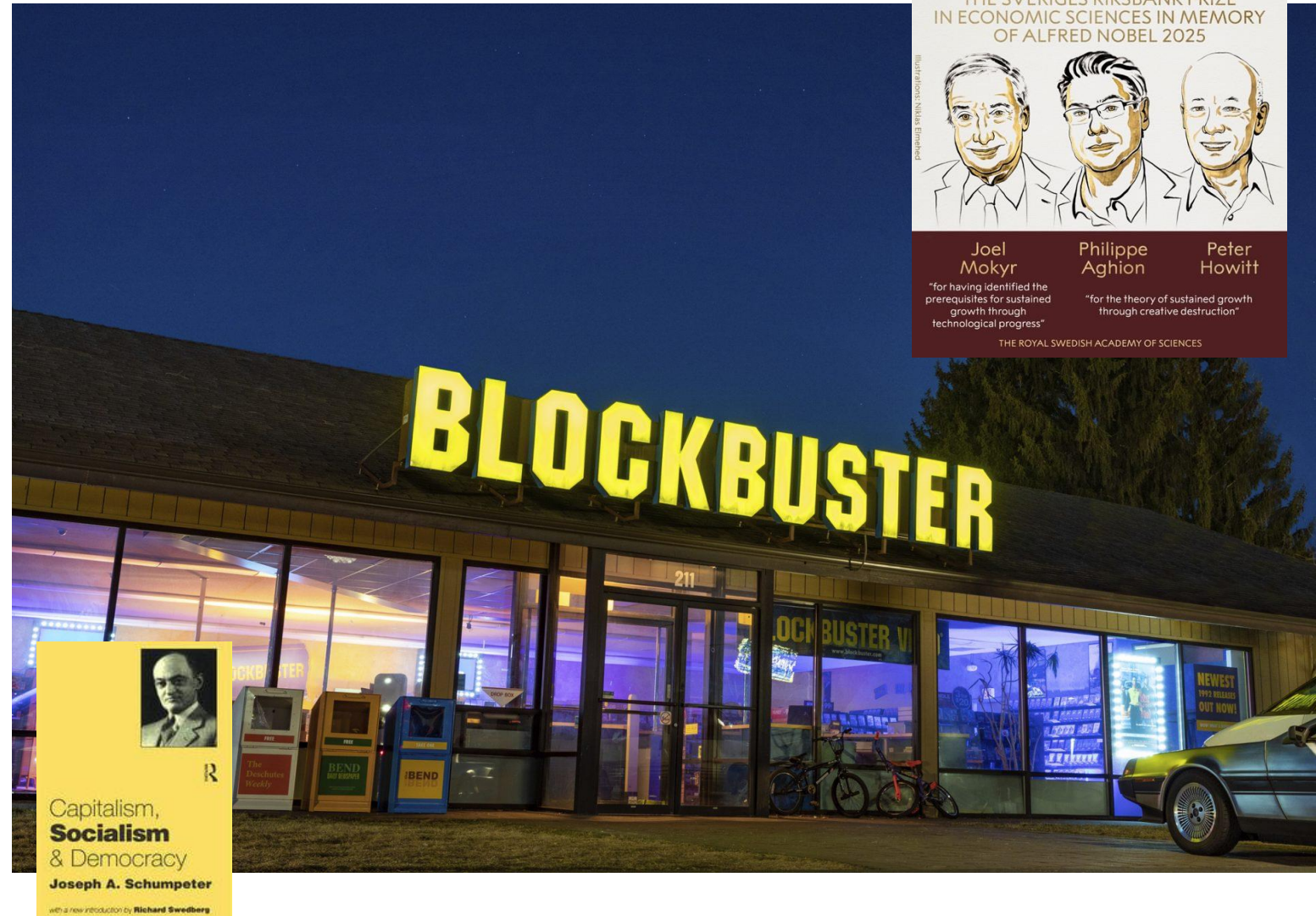
Schumpeter, J. A. (1942). *Capitalism, socialism and democracy*.



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PART II

Individual creativity: insight and ideation

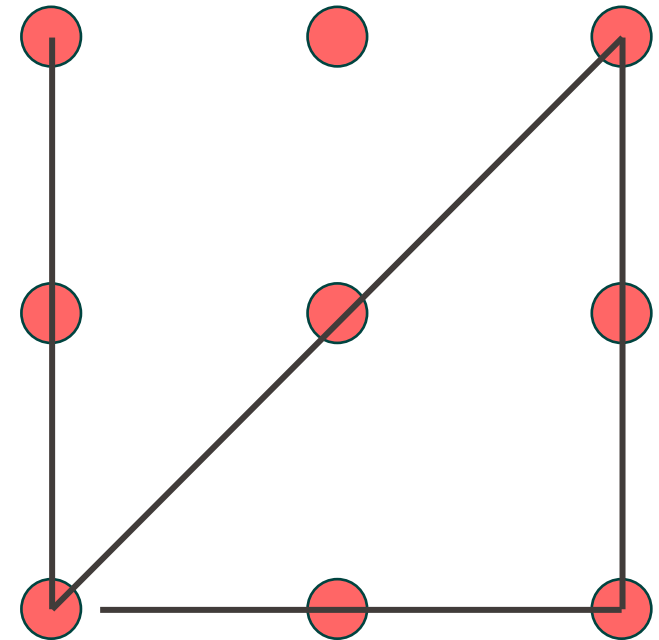
Classic insight problems:

nine-dots

TASK · 2'

Connect 9 dots using **four straight & continuous lines (or fewer)**.

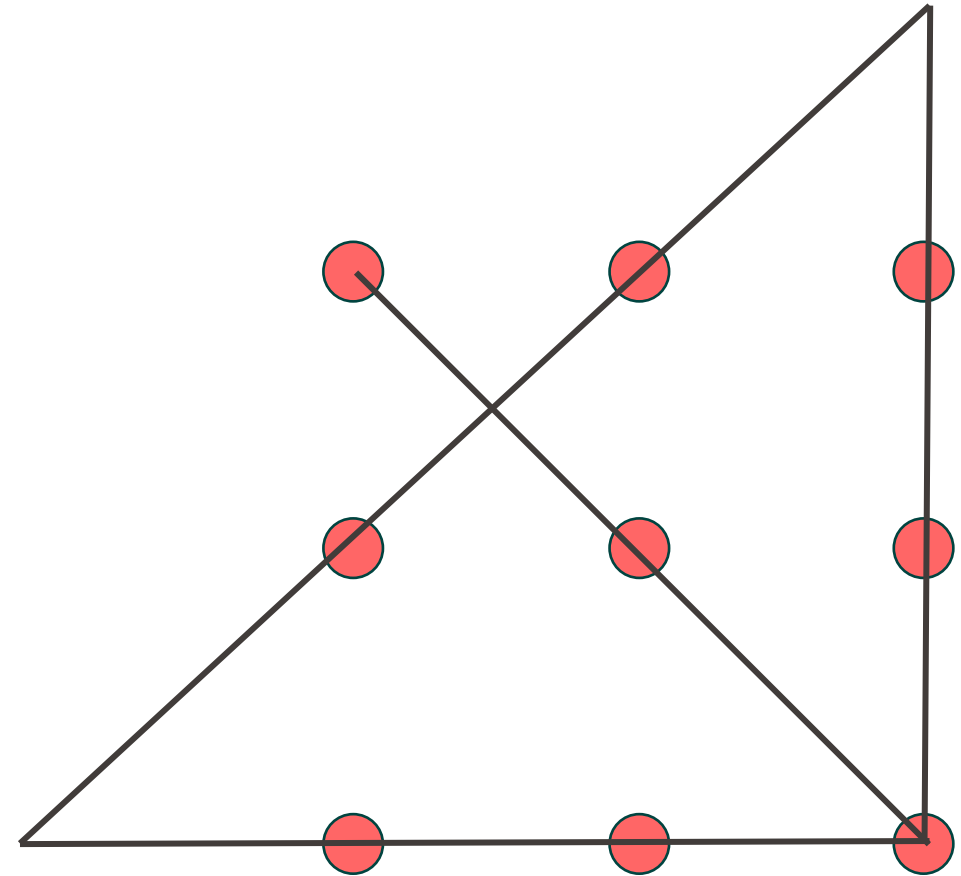
No lifting pen / pencil.



Classic insight problems:

nine-dots

SOLUTION

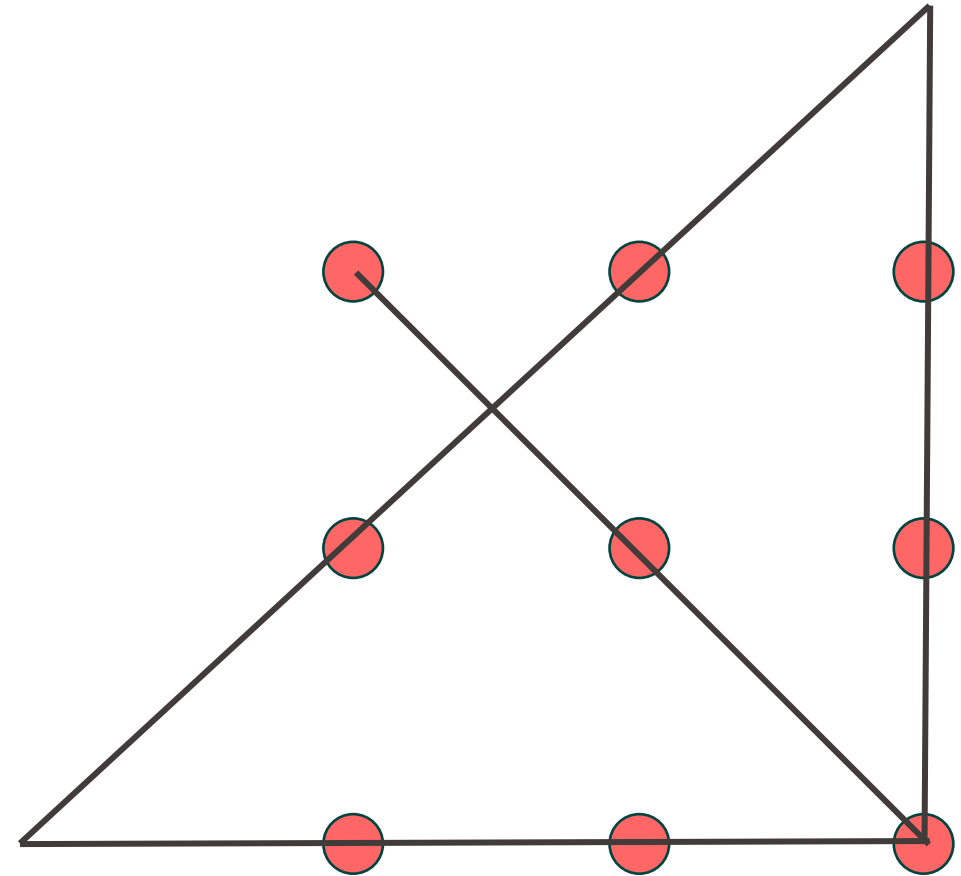


Thinking outside the box

CORE INSIGHTS

Functional fixedness. People struggle to conceive creative solutions because they “fixate” on limited, inappropriate problem-solving angles. (Duncker, 1945)

→ Requires surfacing and violating implicit assumptions of task, representation of input, problem-solving angle; *rule breaking*



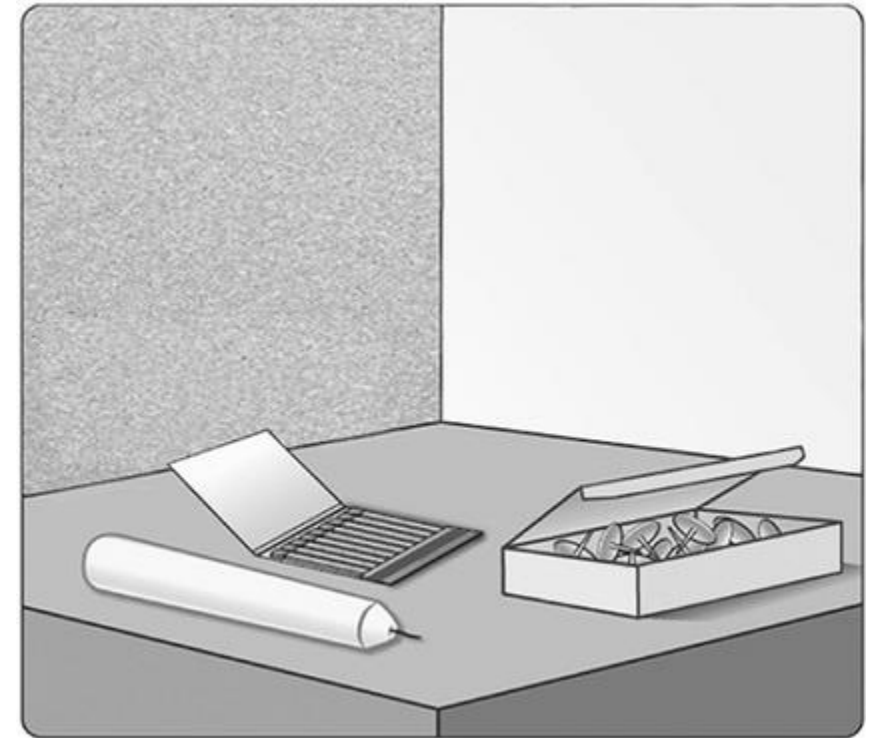
Classic insight problems:

Drucker's candle

TASK • 2'

Items: small candle, box of matches, box of thumbtacks, table, cardboard wall

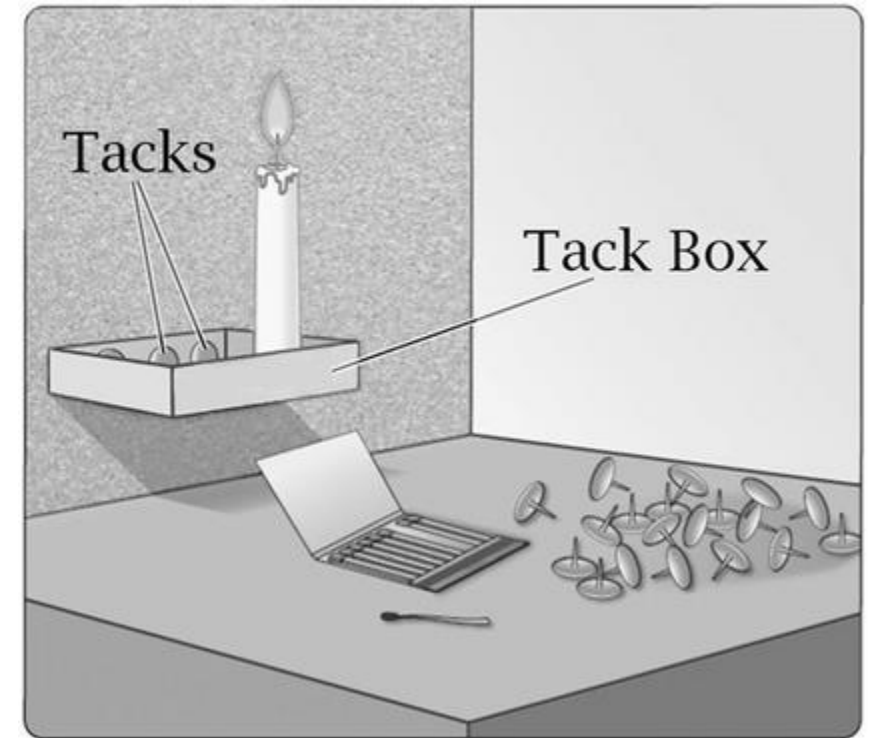
Task: Using only these items, **fix and light** the candle on the wall, so that no wax drips on the table below



Classic insight problems:

Drucker's candle

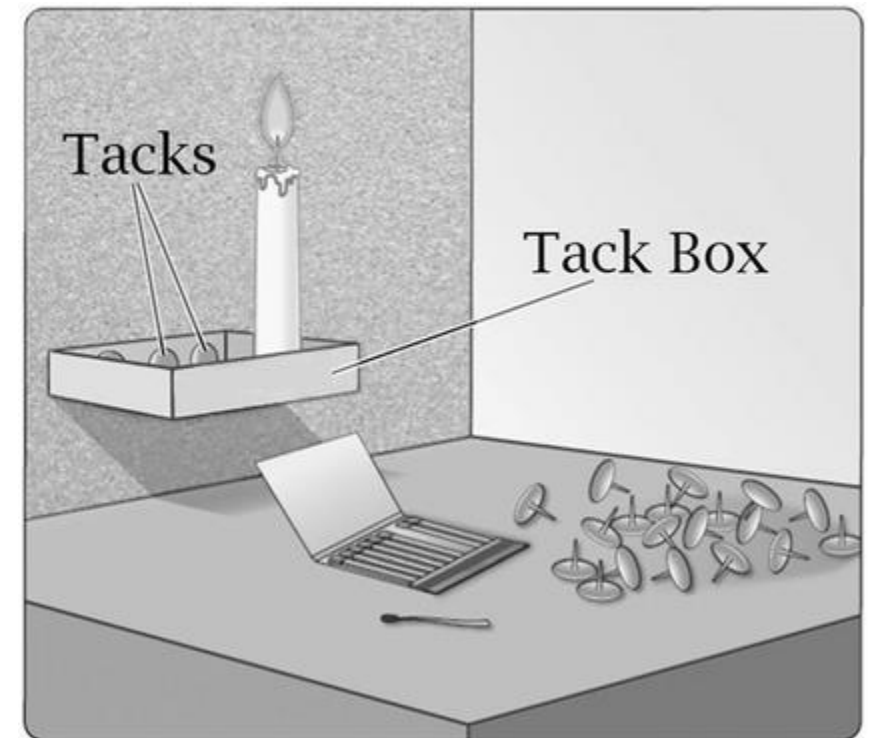
SOLUTION



Heuristics

CORE INSIGHTS

People with professional experience in idea generation rely on **implicit and explicit heuristics** (e.g., recombination, decomposition, delinking and relinking; SCAMPER)



Classic ideation task:

Alternative Uses Tests

TASK · 2'

Task: list as many different and distinct uses for a Coca Cola bottle as you can. One idea per line.



Classic ideation task:

Alternative Uses Tests



Divergent thinking

Ability to generate multiple solutions in response to a given stimulus or problem.

It contrasts with other processes associated to creativity (e.g., convergent thinking, applying criteria, standards, and logics based on their prior knowledge and experience)

Fluency

Number of distinct ideas or solutions



Originality

The relative infrequency of the ideas



Elaboration

The degree of verbal or visual details



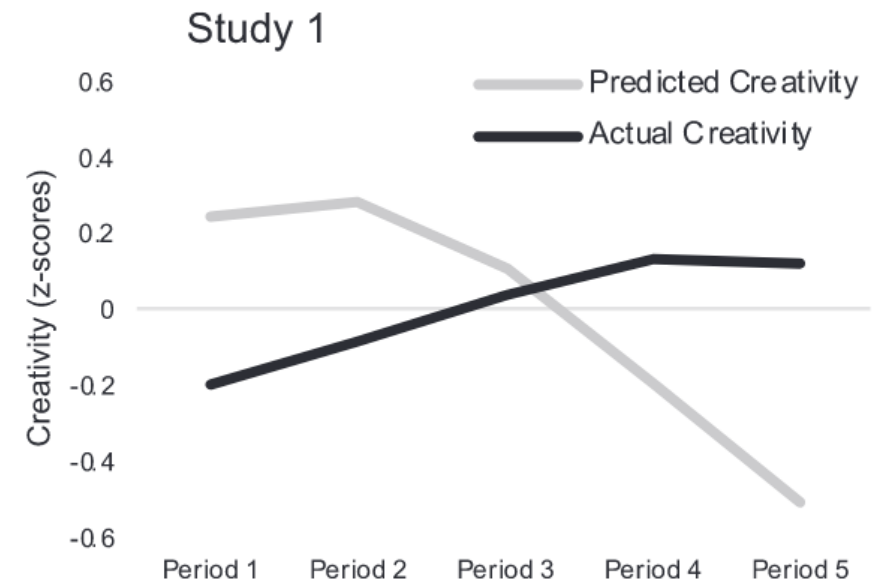
Flexibility

The 'breadth' of categories considered

Divergent thinking playbook

SCIENCE-BASED TACTICS

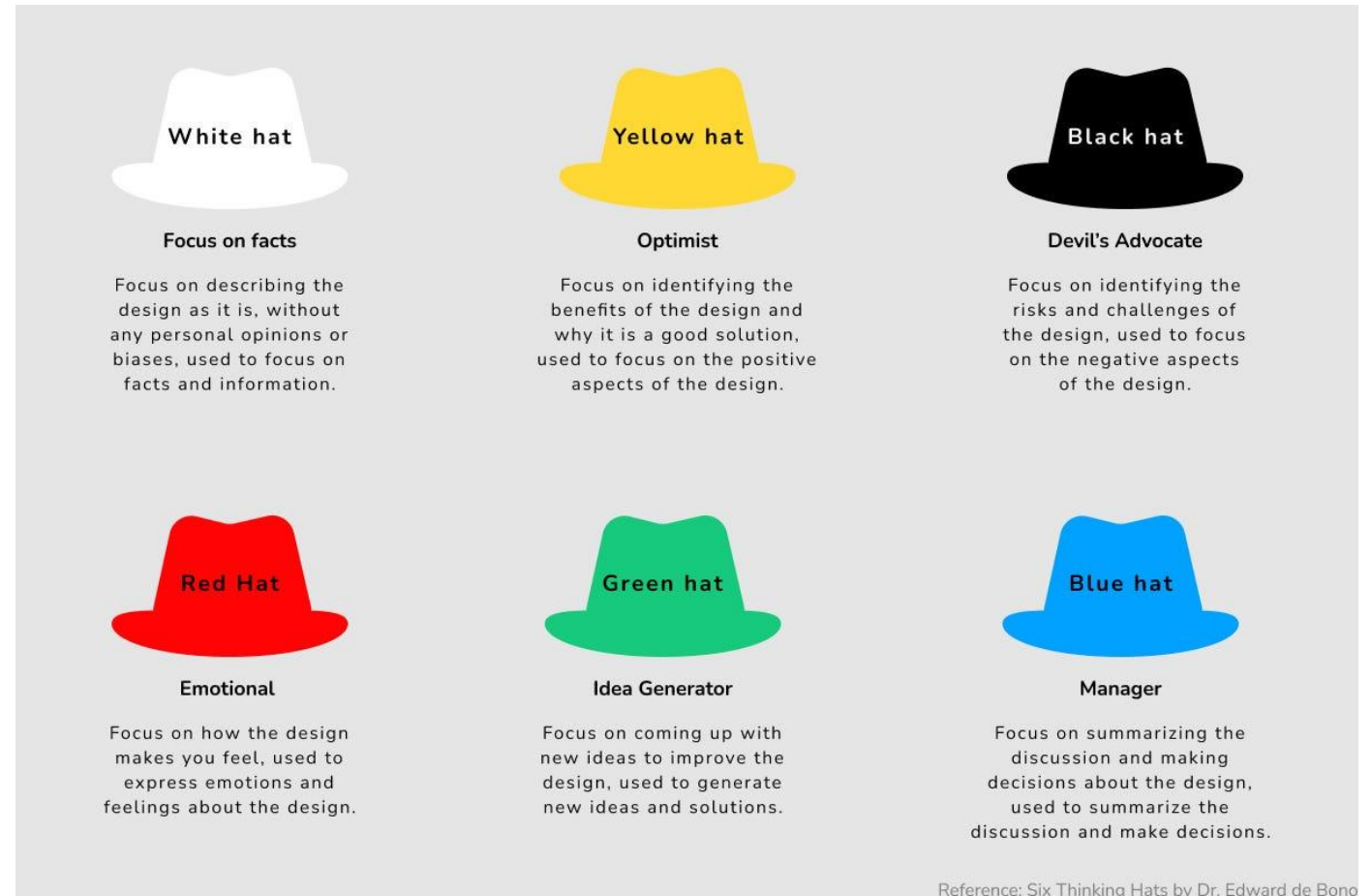
Persistence (and incubation). People consistently underestimated how productive they would be while persisting (the *creative cliff illusion* – Lucas & Nordgren, 2015; 2020). Creativity remains constant or improves across an ideation session.



Divergent thinking playbook

SCIENCE-BASED TACTICS

Psychological distance & perspective taking. As an object becomes more distant (e.g., temporal, spatial, social, hypothetical), our representation becomes more abstract, w/ focus on central, goal-relevant features (vs. context-specific, incidental features).



Divergent thinking playbook

SCIENCE-BASED TACTICS

Creative self-efficacy & selective retention. We are less likely to consider and voice original ideas when we do not consider ourselves to be creative, creative job requirements are high, and there is strong evaluative apprehension.



PART III

Collective creativity

Introduce & connect



Abdelkefi	Ons	9
Abdelmoula	Melik Cyril	10
Alamanos	Panagiotis	1
Alami-Sounni	Kenza Margaux	11
Albertini	Marco	2
Antonou	Jiri	12
Baiges Trilla	Roger	11
Barbanti	Enea	3
Besnard	Mayeul Patrice	12
Boned Ferrer	Sergi	4
Chen	Guanyu	5
Corbière	Clémence Priscilla Marie	5
Cunningham	Sophie Louise	6
Darbellay	Jean-Théo Enrico Valéry	7
Demir	Ayça	1
Dimitrova	Verica	6
Djordjevic	Luka	2
Dutruy	Eloi	3
Egri	Ecenaz	4
El Kasmi	Rania	7

Fata	Jean Marc	5
Ferrari	Paolo	8
Fontaine	Nicolas	6
Foss	Albert	9
Freiberga	Melita	7
Glaser	Philipp	8
Glauser	Loic	9
Gliga	Tatiana	10
Glück	Finn	10
Gonzalez Perez	Valeria	1
Gorgiev	David	2
Grover	Tanmay	3
Hajipour	Pariya	4
Krasna	Aljaz	5
Lefevre	Paul Antoine Marie	6
Maillard	Nathan	7
Malinowski	Stanislaw Piotr	11
Martins Piñon	Lois	8
Mathieux	Gabriel Hugo	11
Menegalli-Boggelli	Patrick	12

Messin-Roizard	Clélia	1
Mikou	Ghita	2
Modenko	Anhelina	3
Moudden	Ismail	9
Okura	Ko	10
Pavan	Giovanni	11
Qi	Yifei	12
Richtr	Jan	4
Robert Halpern	Luis	5
Romdhane	Mohamed Khalil	6
Romero Diaz	Jacobo	12
Sampaio	Alexandre Mattheus	7
Sharma	Tinku	8
Shtilkind	Maksim	8
Silvestre	Jules Pierre Ulysse	9
Tekaya	Ismael	10
Widmer	Yves Denis	1
Wiletsky	Abraham Fischer	2
Wülser	Marc-Antoine	3
Würgler	Raphael	4

Intros (5')

Who am I

What I am up to
these days

Why I am here

Brainstorming

Foresight task

TASK • 5'

Task: list as many different and distinct **future-shaping trends** – macro-changes that may grow in importance in the coming years. Quantity over detail. **Record every idea.** No evaluation and hold judgment.

Brainstorming

Foresight task

Political

Economic

Social

Technological

Legal

Environmental

Why brainstorming fails

CORE INSIGHTS

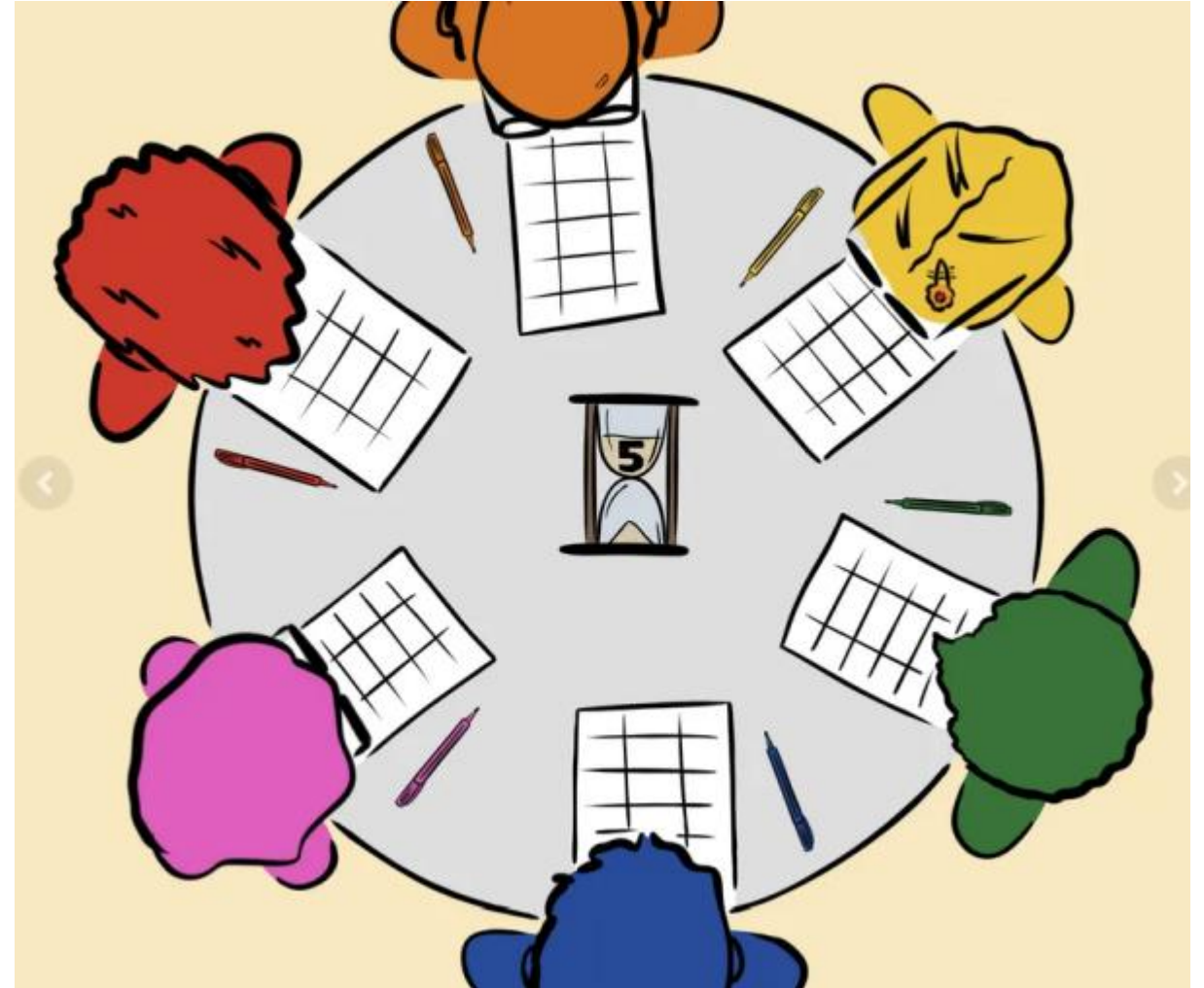
Interactive brainstorming underperforms nominal groups in number, variance and originality of ideas generated due to i) production blocking; ii) evaluation apprehension; iii) free riding; iv) groupthink and fixation / anchoring effects. But they are more fun ;)



A collective creativity playbook

CORE INSIGHTS

Hybrid processes (formal). Better creativity performance when individual ideation and evaluation happens before / during group interaction (e.g., brainwriting “6-3-5” method)



A collective creativity playbook

CORE INSIGHTS

Collective norms and psychological safety. Creativity is supported by the shared belief that is safe to voice ideas, take risks, without fear of negative evaluation. Welcome others' input. Introduce novelty criteria in evaluation to make space for unorthodox, uncertain ideas. Accept divergent interpretations of value.



Brainstorming

Foresight task

Political

Economic

Social

Technological

Legal

Environmental

TASK · 25'

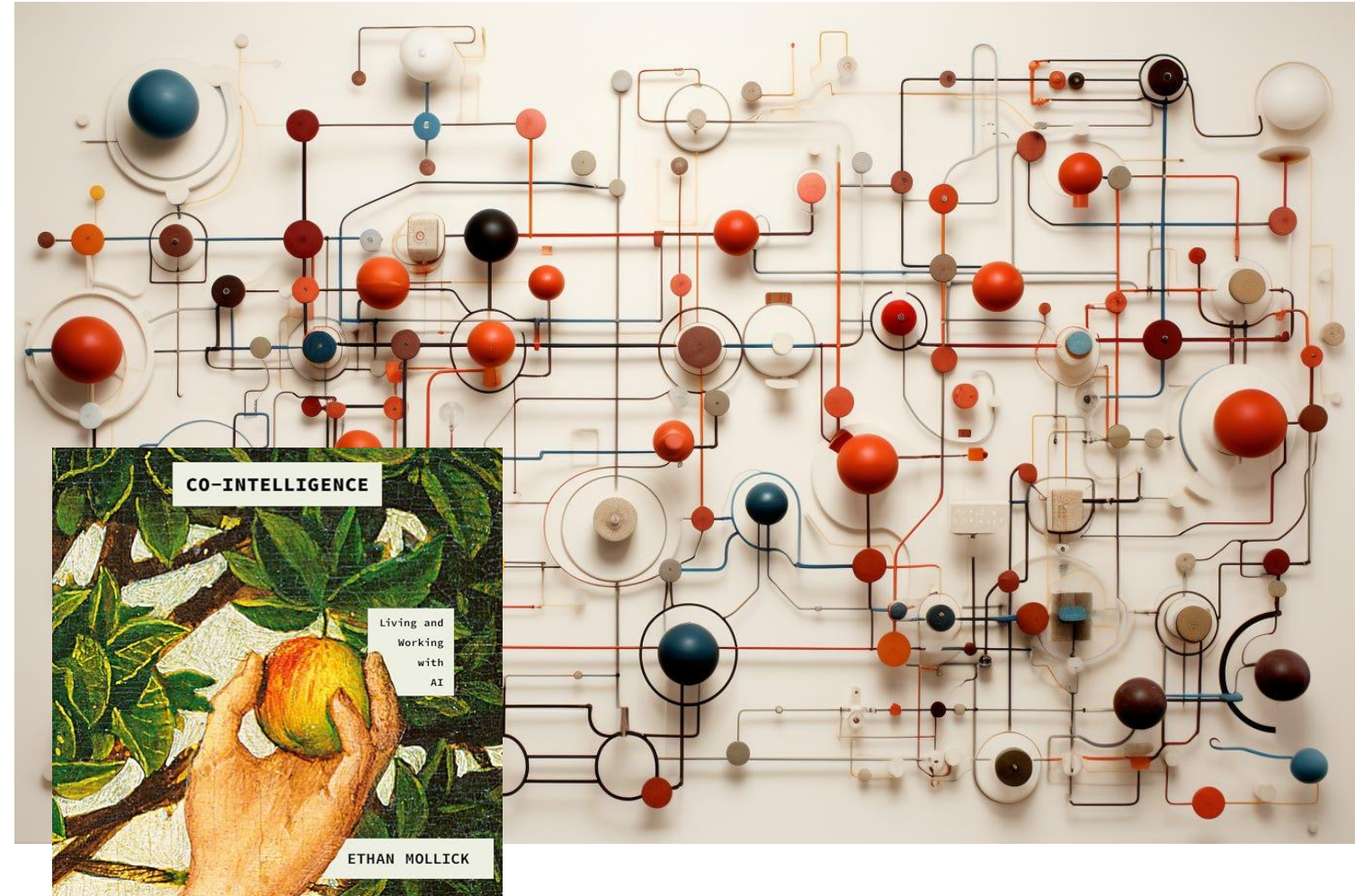
Brainwriting (6-3-5)

6 participants write 3 ideas in 5 minutes over 6 rounds

PART IV

Human-AI co-creation

Should we automate creativity?



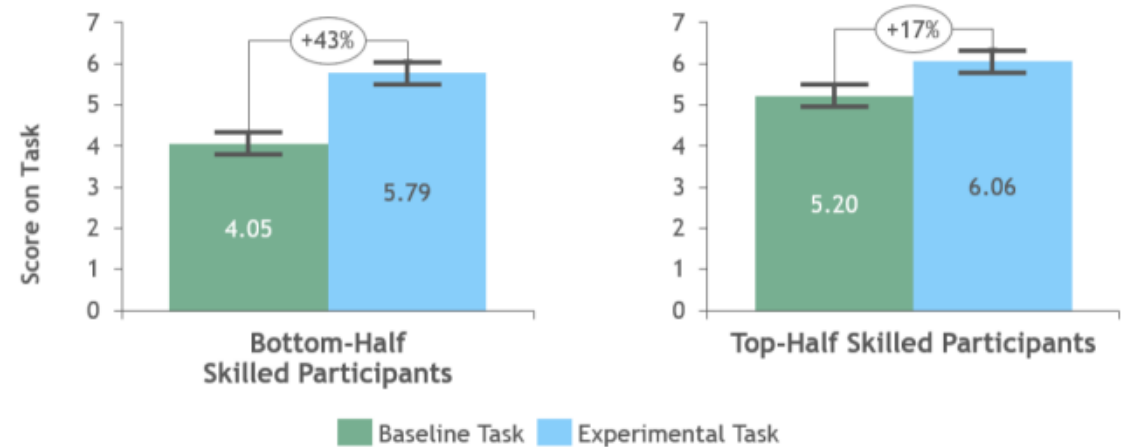
Should we automate creativity?



Who will generate
higher quality and
more creative ideas?
With or without LLM?

Should we automate creativity?

Who will generate higher quality and more creative ideas?
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Should we automate creativity?

Who will generate higher quality and more creative ideas? With or without LLM?

Article

<https://doi.org/10.1038/s41562-024-01953-1>

An empirical investigation of the impact of ChatGPT on creativity

Received: 11 September 2023

Accepted: 10 July 2024

Published online: 12 August 2024

 Check for updates

Byung Cheol Lee ^{1,3}✉ & Jaeyeon (Jae) Chung ^{2,3}

This paper investigates the potential of ChatGPT for helping humans tackle problems that require creativity. Across five experiments, we asked participants to use ChatGPT (GPT-3.5) to generate creative ideas for various everyday and innovation-related problems, including choosing a creative gift for a teenager, making a toy, repurposing unused items and designing an innovative dining table. We found that using ChatGPT increased the creativity of the generated ideas compared with not using any technology or using a conventional Web search (Google). This effect remained robust regardless of whether the problem required consideration of many (versus few) constraints and whether it was viewed as requiring empathetic concern. Furthermore, ChatGPT was most effective at generating incrementally (versus radically) new ideas. Process evidence suggests that the positive influence of ChatGPT can be attributed to its capability to combine remotely related concepts into a cohesive form, leading to a more articulate presentation of ideas.

Doshi, A. R., & Hauser, O. P. (2024). Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28).

Dell'Acqua, F., et al., (2023). Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. *Working Paper*.

Lee, B. C., & Chung, J. (2024). An empirical investigation of the impact of ChatGPT on creativity. *Nature Human Behaviour*, 8(10), 1906-1914.

Should we automate creativity?

Who will generate higher quality and more creative ideas? With or without LLM?

Generative AI enhances individual creativity but reduces the collective diversity of novel content

ANIL R. DOSHI AND OLIVER P. HAUSER [Authors Info & Affiliations](#)

SCIENCE ADVANCES • 12 Jul 2024 • Vol 10, Issue 28 • DOI: 10.1126/sciadv.adn5290

94,097 145



Abstract

Creativity is core to being human. Generative artificial intelligence (AI)—including powerful large language models (LLMs)—holds promise for humans to be more creative by offering new ideas, or less creative by anchoring on generative AI ideas. We study the causal impact of generative AI ideas on the production of short stories in an online experiment where some writers obtained story ideas from an LLM. We find that access to generative AI ideas causes stories to be evaluated as more creative, better written, and more enjoyable, especially among less creative writers. However, generative AI-enabled stories are more similar to each other than stories by humans alone. These results point to an increase in individual creativity at the risk of losing collective novelty. This dynamic resembles a social dilemma: With generative AI, writers are individually better off, but collectively a narrower scope of novel content is produced. Our results have implications for researchers, policy-makers, and practitioners interested in bolstering creativity.

Human-AI co-creation



Human-AI co-creation

Prompting Diverse Ideas: Increasing AI Idea Variance

38 Pages • Posted: 12 Feb 2024

[Lennart Meincke](#)

University of Pennsylvania; The Wharton School

[Ethan R. Mollick](#)

University of Pennsylvania - Wharton School

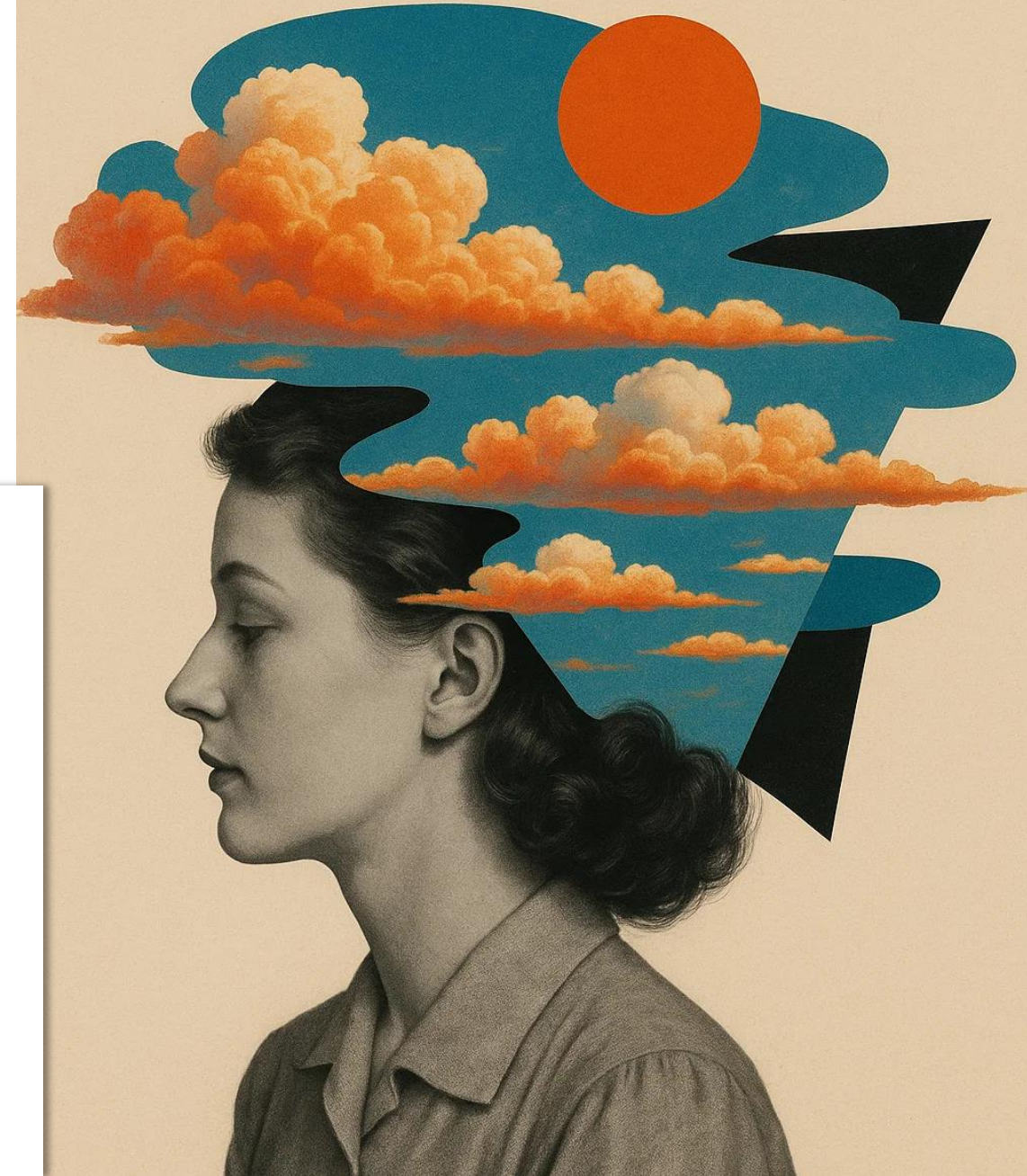
[Christian Terwiesch](#)

University of Pennsylvania - Operations & Information Management Department

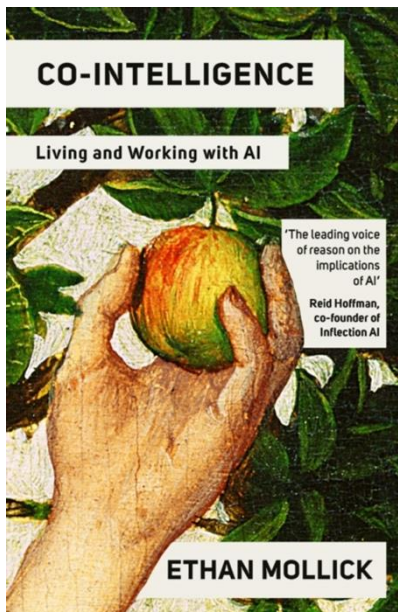
Date Written: January 27, 2024

Abstract

Unlike routine tasks where consistency is prized, in creativity and innovation the goal is to create a diverse set of ideas. This paper delves into the burgeoning interest in employing Artificial Intelligence (AI) to enhance the productivity and quality of the idea generation process. While previous studies have found that the average quality of AI ideas is quite high, prior research also has pointed to the inability of AI-based brainstorming to create sufficient dispersion of ideas, which limits novelty and the quality of the overall best idea. Our research investigates methods to increase the dispersion in AI-generated ideas. Using GPT-4, we explore the effect of different prompting methods on Cosine Similarity, the number of unique ideas, and the speed with which the idea space gets exhausted. We do this in the domain of developing a new product development for college students, priced under \$50. In this context, we find that (1) pools of ideas generated by GPT-4 with various plausible prompts are less diverse than ideas generated by groups of human subjects (2) the diversity of AI generated ideas can be substantially improved using prompt engineering (3) Chain-of-Thought (CoT) prompting leads to the highest diversity of ideas of all prompts we evaluated and was able to come close to what is achieved by groups of human subjects. It also was capable of generating the highest number of unique ideas of any prompt we studied.



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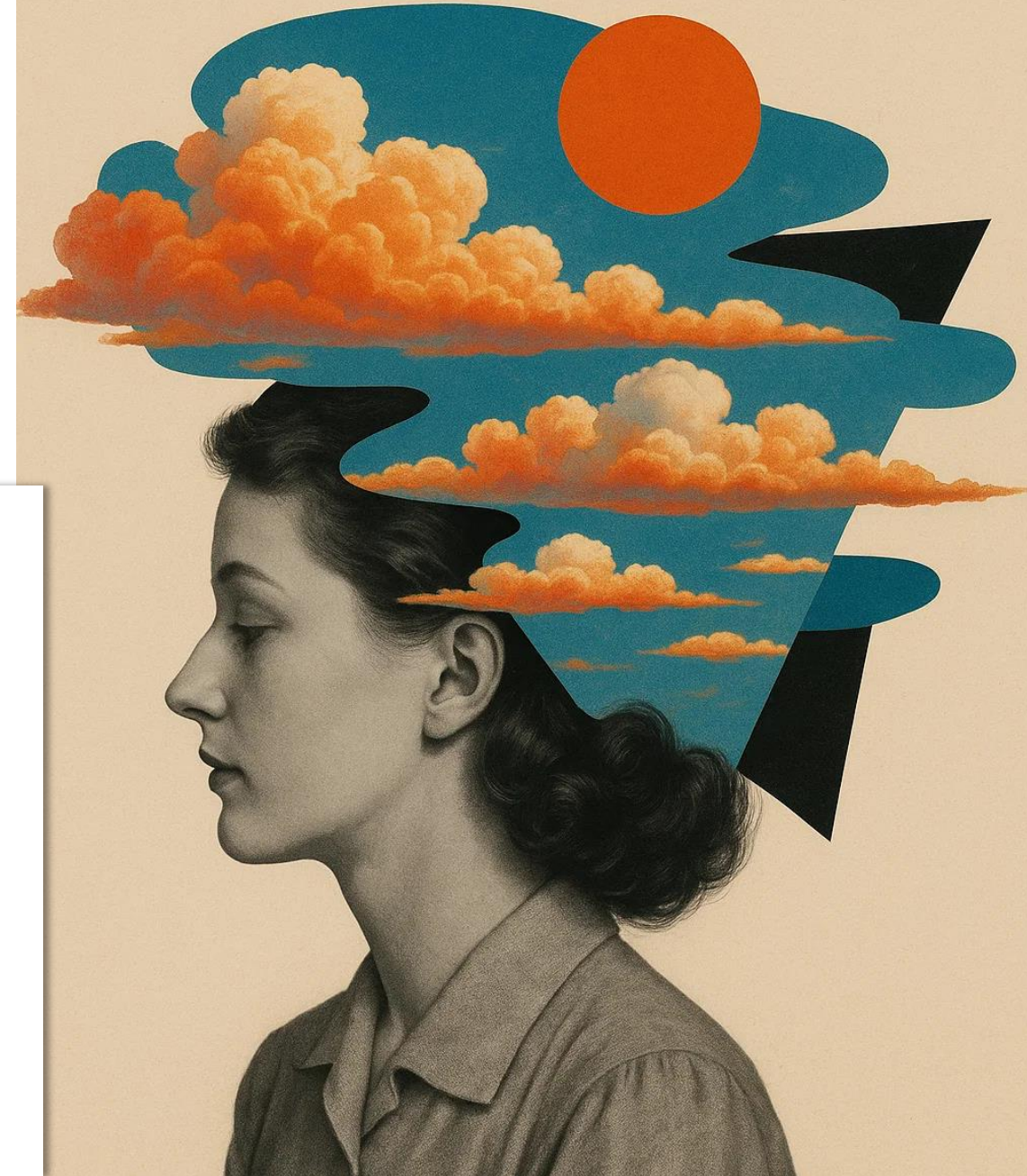
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PART V

Next steps

About the venture concepts

Opportunity set (20%) · Individual

Visual posters of at least two of the most promising venture concepts you identified from broad set of market opportunities + 60-second elevator pitch of the one you intend to pursue.

Share .pdf file via **Discord** by **noon of the day before Session 5.**

About the venture concepts

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Your venture idea.
In ten words – or less.

Problem. A defined user segment faces a costly, common pain point; willingness to pay is established, and existing options have structural gaps that limit outcomes. **Promise.** We apply proven practices to deliver measurable gains on the metrics that matter. Our method builds on established work and a new approach that supports durable competitive advantage. **Proof.** Team experience aligns with the tech / industry; technical reports show performance above typical benchmarks. **Why now.** Regulation changes and maturing infrastructure reduce previous barriers to adoption and support near-term execution.

You

Name, Surname

Study program

keyword • keyword • keyword

keyword • keyword • keyword

About the venture concepts

Opportunity set (20%) · Individual

Visual posters of at least two of the most promising venture concepts you identified from broad set of market opportunities + 60-second elevator pitch of the one you intend to pursue.

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I will look for breadth (opportunity set), potential / challenge (next week), delivery (pitch)



**Your venture idea.
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You

Name, Surname

Study program

keyword · keyword · keyword

keyword · keyword · keyword

An example



Before → After → Signals



Pierre Coutaz
Microengineering
watchmaking

An example



Before → After → Signals



Pierre Coutaz
Microengineering
watchmaking

■ TECHNOLOGY VENTURES



Before → After → Signals



Pierre Coutaz
Microengineering
watchmaking

■ TECHNOLOGY VENTURES

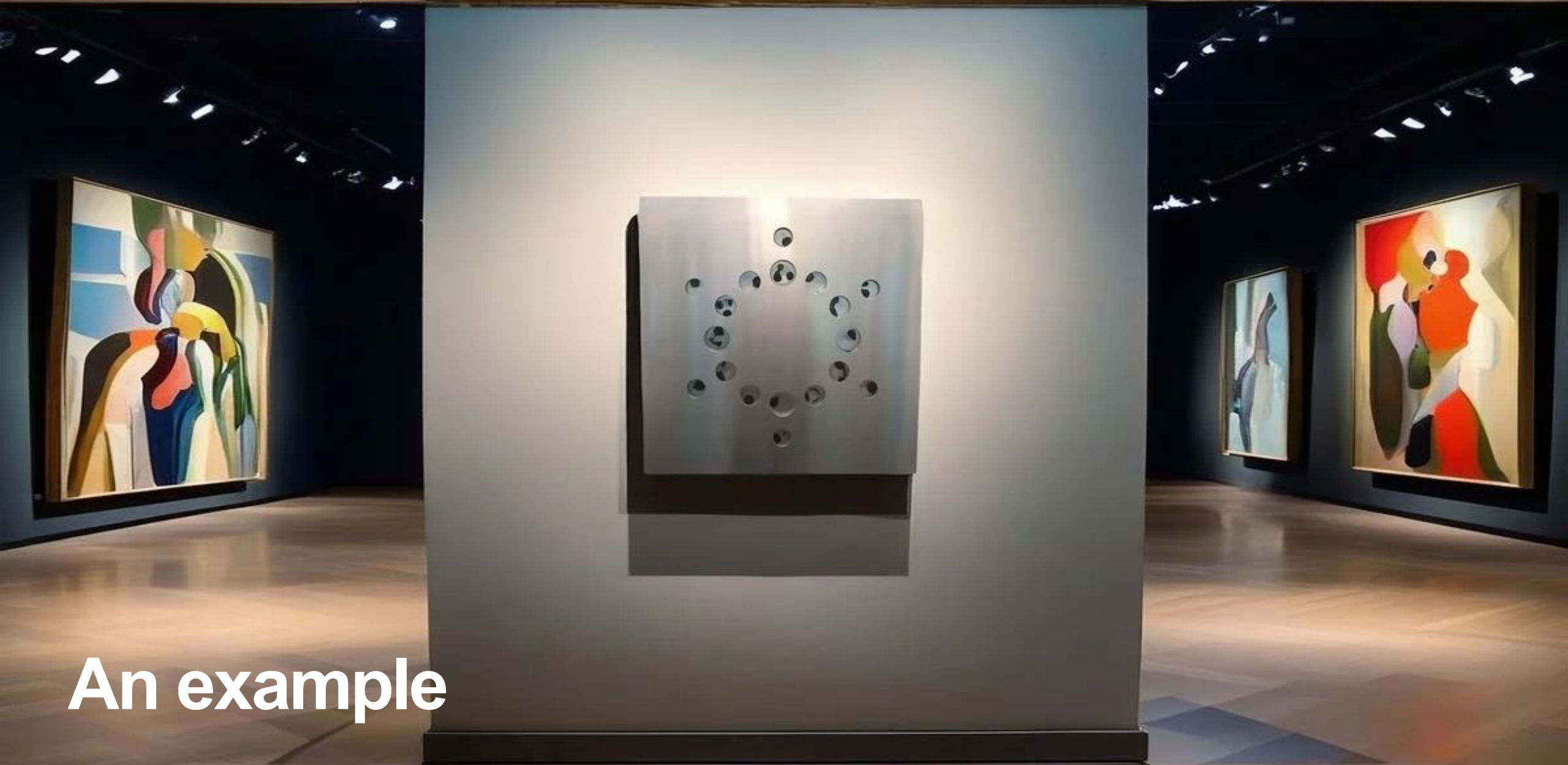


Before → After → Signals



Pierre Coutaz
Microengineering
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■ TECHNOLOGY VENTURES



An example